

Supplementary Files

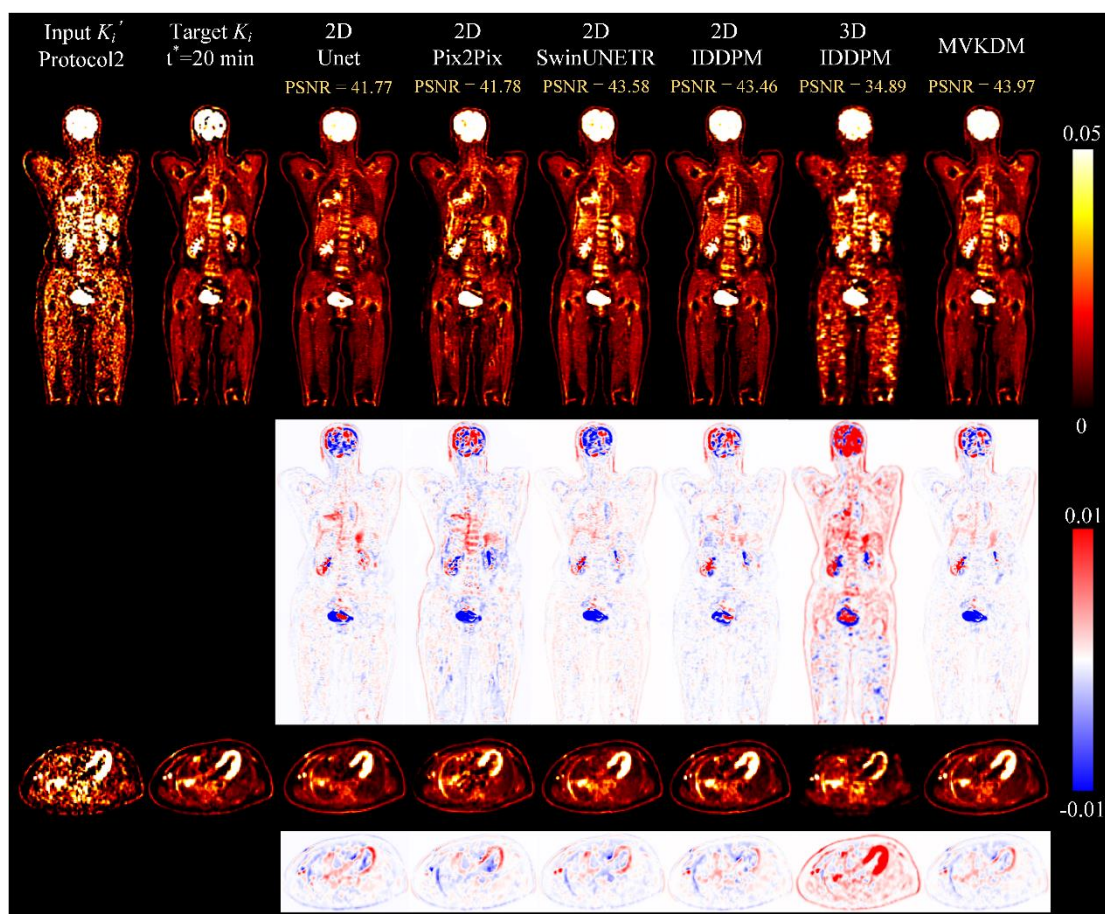


Fig. S1: Error maps between the synthetic and reference images by different methods under protocol2 (40-60min p.i.).

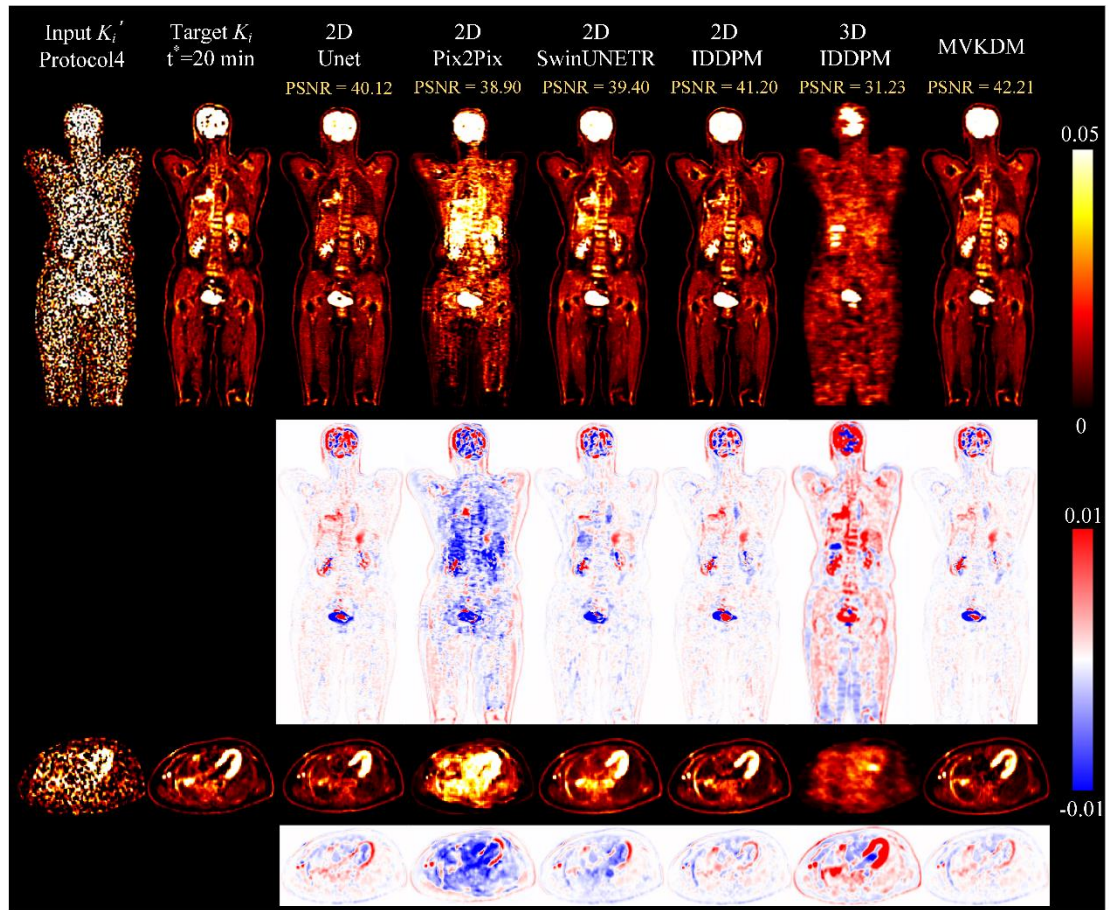


Fig. S2: Error maps between the synthetic and reference images by different methods under protocol3 (50-60min p.i.).

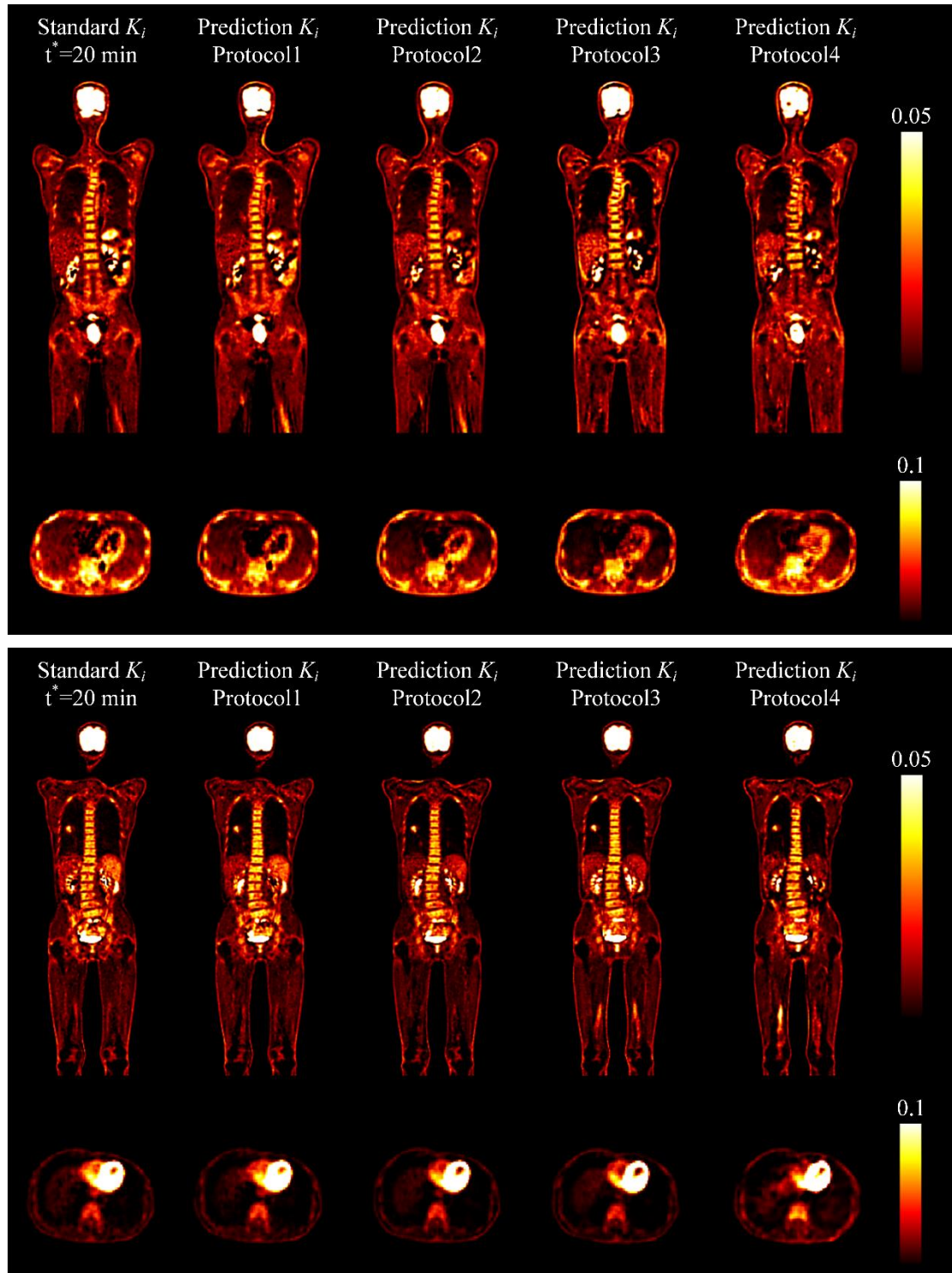


Fig. S3: Additional examples of synthesized K_i images under four shortened protocols (top: patient #1; bottom: patient #2).

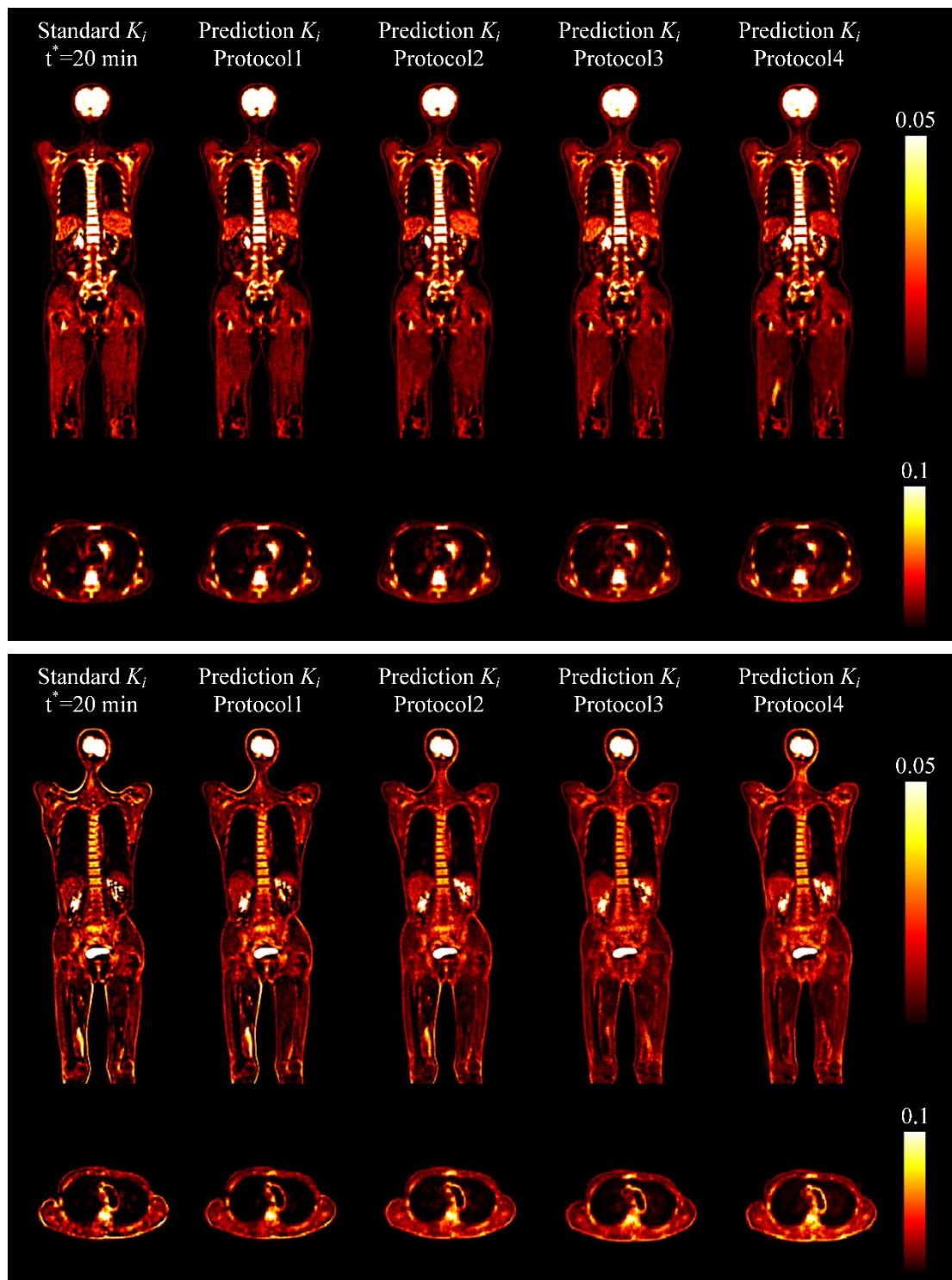


Fig. S4: Additional examples of synthesized K_i images under four shortened protocols (top: patient #3; bottom: patient #4).

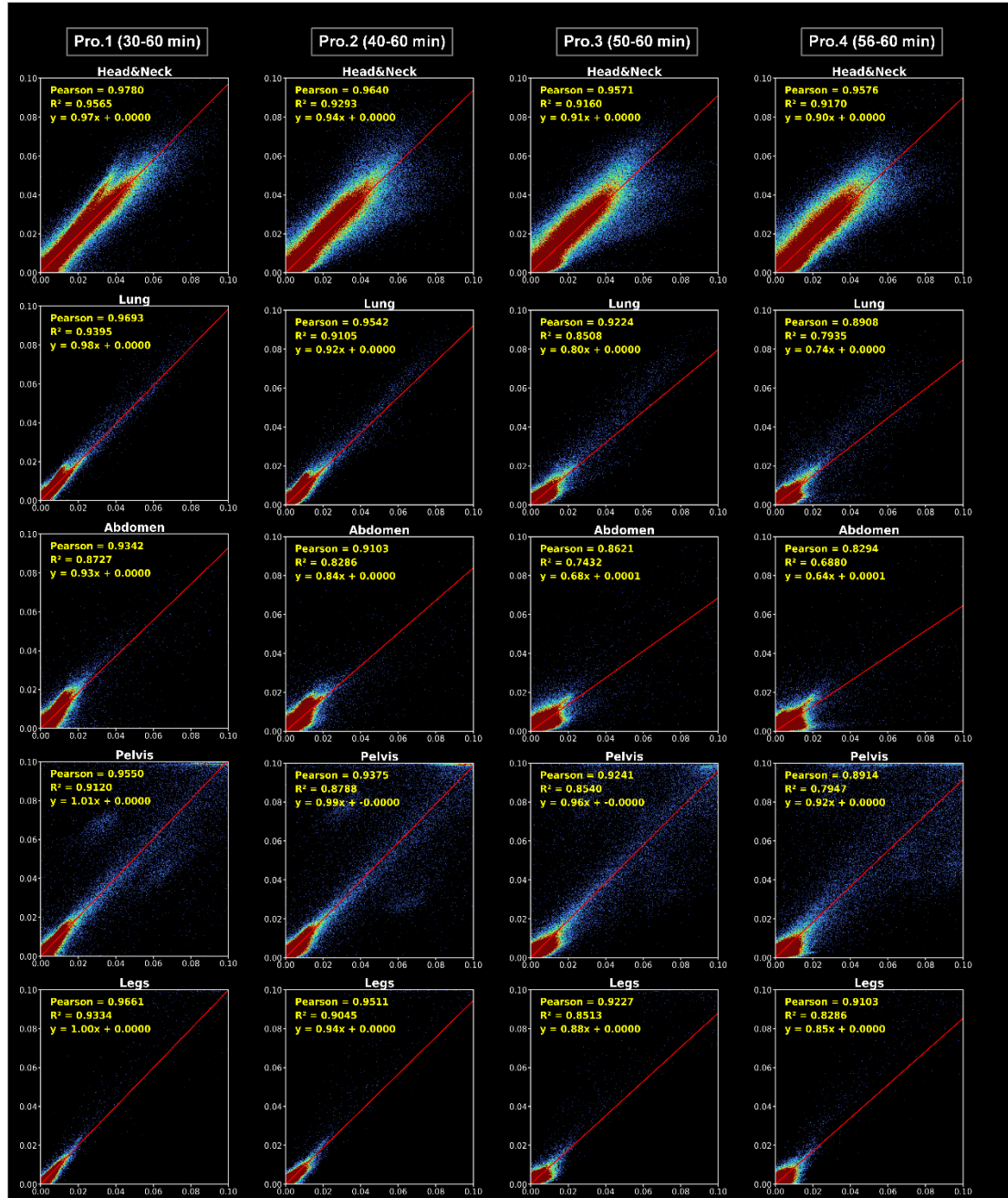


Fig. S5: Joint histogram analysis for images synthesized by the MVKDM method across five regions (Head & Neck, Lung, Abdomen, Pelvis, and Legs) under four protocols.

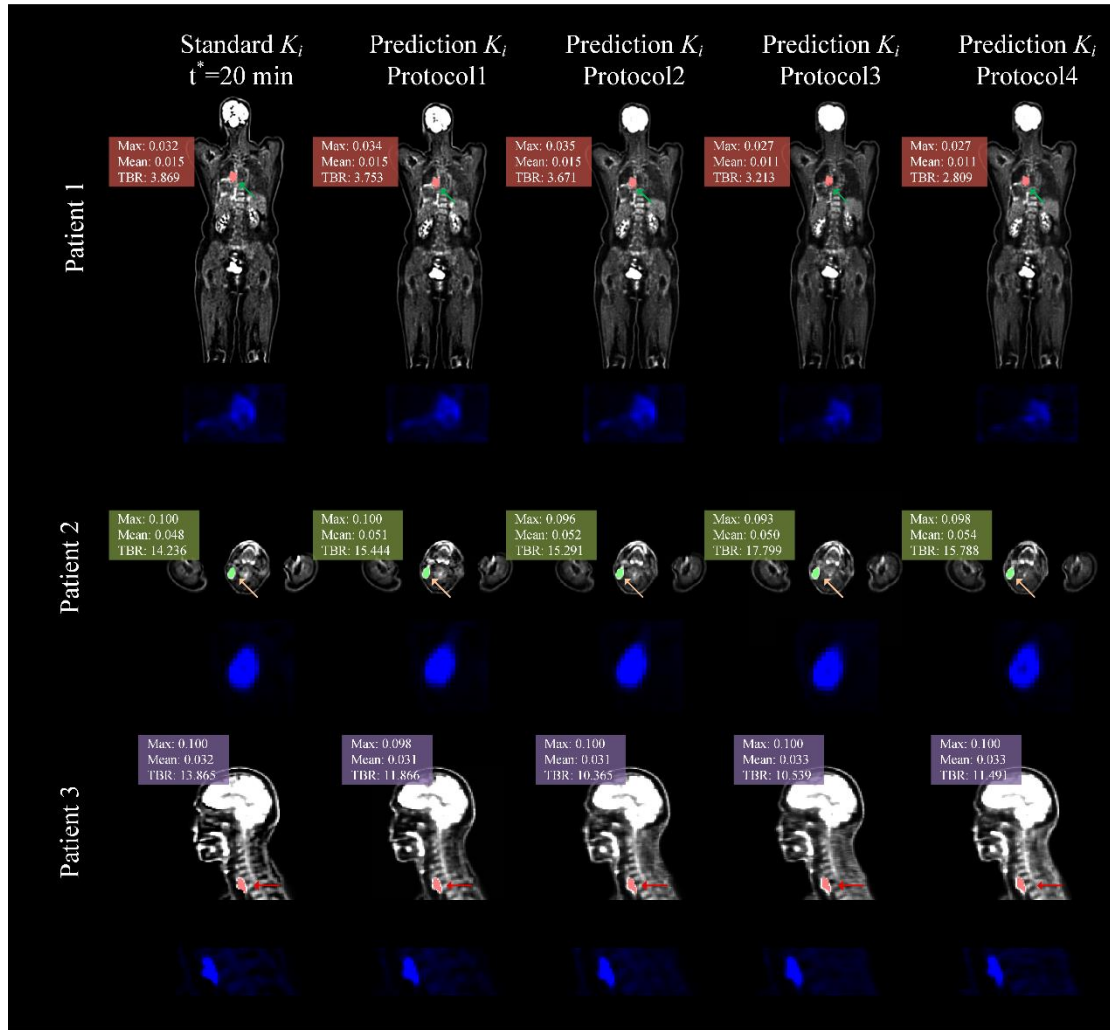


Fig. S6: Representative lesion-level visual comparison of the reference and synthesized K_i images. From top to bottom, the rows correspond to patient #1 (lung cancer), patient #2 (lymph node metastasis), and patient #3 (esophageal cancer). The first column shows the reference K_i image generated from the full dynamic PET scan ($t^* = 20$ min), and the remaining columns show the synthesized K_i images under Protocols 1–4. Colored overlays and arrows indicate the delineated lesion regions. The enlarged insets below each image show zoomed-in tumor-region visualizations. For each lesion, maximum K_i , mean K_i , and tumor-to-background ratio (TBR) using liver as background are reported within the lesion mask.

Table S1: Tumor type distribution of the patient cohort.

Tumor type	Number of patients
Lung cancer	6
Esophageal cancer	2
Liver cancer	2
Nasopharyngeal carcinoma	1
Rectal cancer	1
Renal cancer	1
Tonsillar carcinoma	1
Breast cancer	1
Total	15

Table S2: The absolute bias across six organs under four protocols.

	Brain	Lung	Liver	Spleen	Kidney	Heart
Pro.1	0.0033±0.00 18	0.0003±0.00 01	0.0010±0.00 07	0.0008±0.00 05	0.0024±0.00 07	0.0009±0.00 05
Pro.2	0.0041±0.00 23	0.0004±0.00 09	0.0013±0.00 07	0.0011±0.00 05	0.0032±0.00 11	0.0012±0.00 05
Pro.3	0.0046±0.00 25	0.0005±0.00 02	0.0016±0.00 11	0.0015±0.00 08	0.0037±0.00 13	0.0016±0.00 07
Pro.4	0.0047±0.00 21	0.0005±0.00 02	0.0020±0.00 13	0.0015±0.00 07	0.0037±0.00 13	0.0017±0.00 10

Table S3: Lesion-level quantitative results

Lesion category	Number of lesion regions	Target Mean/Max /TBR	Pro.1 Mean/Max /TBR	Pro.2 Mean/Max /TBR	Pro.3 Mean/Max /TBR	Pro.4 Mean/Max /TBR
Lung cancer	3	0.011/0.026 /3.111	0.011/0.026 /2.830	0.011/0.028 /2.932	0.010/0.025 /2.632	0.009/0.022 /2.358
Recurrent lung cancer	2	0.007/0.016 /2.294	0.007/0.015 /2.224	0.007/0.017 /2.328	0.007/0.017 /1.928	0.007/0.013 /1.164
Esophageal cancer	2	0.030/0.100 /13.527	0.032/0.087 /10.416	0.029/0.075 /7.906	0.028/0.081 /8.365	0.027/0.076 /9.430
Nasopharyngeal carcinoma	1	0.019/0.049 /5.551	0.018/0.046 /5.384	0.022/0.054 /6.552	0.019/0.045 /6.826	0.018/0.041 /5.356
Residual hepatic tumor	2	0.006/0.012 /2.717	0.005/0.008 /1.648	0.004/0.008 /1.468	0.005/0.010 /1.395	0.004/0.009 /1.416
Lymph node metastasis	6	0.019/0.060 /6.363	0.020/0.061 /6.135	0.021/0.055 /6.172	0.019/0.053 /6.173	0.020/0.056 /5.598
Lung metastasis	3	0.008/0.024 /2.807	0.008/0.021 /2.226	0.010/0.029 /2.840	0.010/0.028 /2.509	0.011/0.033 /2.792
Bone metastasis	1	0.019/0.061 /8.843	0.018/0.055 /6.061	0.017/0.054 /6.096	0.019/0.056 /4.981	0.021/0.063 /6.738
Rib metastasis	1	0.013/0.059 /5.584	0.013/0.053 /5.035	0.018/0.065 /6.865	0.021/0.067 /6.914	0.017/0.054 /4.800
Thyroid metastasis	1	0.024/0.079 /10.204	0.023/0.095 /8.877	0.027/0.091 /10.175	0.028/0.090 /9.461	0.027/0.090 /7.866

Nasopharyngeal metastasis	1	0.026/0.076 /11.156	0.025/0.074 /9.734	0.034/0.080 /12.734	0.029/0.069 /9.626	0.028/0.063 /8.048
Portal vein tumor thrombus	1	0.003/0.011 /1.310	0.003/0.007 /1.000	0.002/0.007 /0.851	0.003/0.007 /0.790	0.003/0.006 /0.806
Presacral soft-tissue mass	1	0.006/0.034 /1.344	0.005/0.028 /1.300	0.005/0.032 /1.281	0.005/0.029 /1.208	0.005/0.029 /1.516

This table presents lesion-level quantitative results grouped by lesion category. Each quantitative cell reports mean K_i , maximum K_i , and liver-background tumor-to-background ratio (TBR), respectively. Values were averaged across lesion-level regions of interest within each category, with each region weighted equally.

Table S4: Model complexity and inference efficiency comparison.

	2D Unet	Pix2pix	SwinUNETR	2D IDDPM	3D IDDPM	MVKDM
Parameters	14.54 M	61.67 M	25.14 M	19.39 M	51.98 M	58.16 M
Times/case	12.38 s	12.86 s	16.01 s	301.70 s	432.22 s	449.9 s